

# Yiwei Chen

✉ [chenyiw9@msu.edu](mailto:chenyiw9@msu.edu) | 🌐 [Homepage](#) | [in](#) [LinkedIn](#) | 📞 +1 5172046319

## EDUCATION

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### Michigan State University

East Lansing, MI, USA

- **Ph.D. in Computer Science**

Aug. 2024 -

Advisor: Prof. [Sijia Liu](#)

### Xi'an Jiaotong University

Xi'an, Shaanxi, China

- **M.S. in Computer Science and Technology.** Ranking: 7/173

Sept. 2021 - Jun. 2024

- **B.Eng. in Automation.**

Sept. 2017 - Jun. 2021

Honor Class in Qian Xuesen Honors College.

Ranking: 3/24

- **Special Class for Gifted Young.**

Sept. 2015 - Jun. 2017

A honor program for nationwide selected gifted young students.

## RESEARCH INTERESTS

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- Reasoning, Post-training, Large Language Models, Multi-modal Language Models
- AI Safety, Alignment, Trustworthy Algorithms, Interpretability, Machine Unlearning

## PUBLICATIONS & MANUSCRIPTS

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(\* means Equal Contribution)

- [1] **Y. Chen**\*, S. Pal\*, Y. Zhang, Q. Qu, S. Liu. “Unlearning Isn’t Invisible: Detecting Unlearning Traces in LLMs from Model Outputs.” *The International Conference on Learning Representations (ICLR)*, 2026. [\[Paper\]](#)
- [2] **Y. Chen**\*, Y. Yao\*, Y. Zhang, B. Shen, G. Liu, S. Liu. “Safety Mirage: How Spurious Correlations Undermine VLM Safety Fine-Tuning and Can Be Mitigated by Machine Unlearning.” *The International Conference on Learning Representations (ICLR)*, 2026. [\[Paper\]](#)
- [3] B. Shang\*, **Y. Chen**\*, Y. Zhang, B. Shen, S. Liu. “Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning.” In Submission to ICML’26. [\[Paper\]](#)
- [4] Z. Zhang\*, **Y. Chen**\*, W. Zhang, C. Yan, Q. Zheng, Q. Wang, W. Chen. “Tile Classification Based Viewport Prediction with Multi-modal Fusion Transformer.” *ACM Multimedia (ACM-MM)*, 2023. [\[Paper\]](#)
- [5] Y. Zhang, C. Wang, **Y. Chen**, C. Fan, J. Jia, S. Liu. The Fragile Truth of Saliency: Improving LLM Input Attribution via Attention Bias Optimization. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025, Spotlight (3.2% of 21575 submissions). [\[Paper\]](#)

## INTERNSHIPS

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Cisco, Research Intern

San Francisco, US

Mentor: [Lichi Li](#)

May 2025 - Oct. 2025

Research Project: Reasoning Large Language Models for CVE Exploit Generation in Cybersecurity

- *Exploit Generation*: Developed a benchmark with 6 input levels and 7 evaluation metrics, providing the first systematic framework for evaluating LLMs on exploit generation.
- *LLM Reasoning*: Collected a large-scale dataset of CVE-exploit pairs and fine-tuned Qwen3-8B using SFT + GRPO, yielding significant performance gains over SOTA LLMs (GPT, Claude, Qwen, etc).

# RESEARCH EXPERIENCE

**OPTML, Michigan State University**  
Research Assistant. Advisor: Prof. [Sijia Liu](#)

*East Lansing, US*  
Aug. 2024 -

- *MLLM Reasoning*: Existing improvements in mathematical reasoning for MLLMs are largely transferred from LLMs, with insufficient utilization of visual information during training. Propose visual-biased GRPO training that enhances visual grounding and improves reasoning performance.
- *LLM Unlearning* [1]: Investigated how machine unlearning leaves persistent, detectable “fingerprints” in both outputs and internal activations. Designed classifiers could identifying unlearned models, revealing new risks of reverse-engineering forgotten information.
- *MLLM Unlearning* [2]: Conventional safety fine-tuning of MLLMs suffers from a “safety mirage” caused by training bias, leading to spurious correlations and over-rejections under one-word attacks. Unlearning algorithms effectively remove harmful content and mitigate these issues.
- *Backdoor Unlearning* [3]: Demonstrated that LLM unlearning can be compromised through attention-sink-guided backdoor unlearning, where triggers placed at attention sinks enable models to recover forgotten knowledge while maintaining normal behavior without triggers.

# AWARDS & HONORS

## SCHOLARSHIPS

- Michigan State University Travel Award Oct. 2025
- Outstanding Master Graduate, Xi'an Jiaotong University (Top 5% of all graduates) Jun. 2024
- Outstanding Bachelor Graduate, Xi'an Jiaotong University (Top 5% of all graduates) Jun. 2021
- Outstanding Freshman Scholarship, Xi'an Jiaotong University (Top 10%) Nov. 2021
- First Prize Scholarship, Xi'an Jiaotong University (Top 10%) Nov. 2016-2019, 2022-2023

## COMPETITIONS

- Meritorious Winner in Mathematical Contest in Modelling (top 8% in 25370 teams) Mar. 2019
- The First Prize in Shaanxi of China Undergraduate Mathematical Contest in Modeling Oct. 2018

# ACADEMIC SERVICE

## Conference Reviewer

- The International Conference on Learning Representations (ICLR 2025-2026)
- IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2025-2026)
- ACM International Conference on Multimedia (ACM-MM 2023-2024)
- The International Joint Conference on Neural Networks (IJCNN 2025)

# LANGUAGE & SKILLS

|                       |                                                                         |
|-----------------------|-------------------------------------------------------------------------|
| Language              | Mandarin (Native)                                                       |
|                       | English (TOEFL iBT 103, Reading 28 Listening 28 Speaking 25 Writing 22) |
| Programming           | Python, C/C++, Bash, LaTeX, MATLAB, HTML/CSS                            |
| Model Frameworks      | LLMs (e.g., Llama, Qwen), MLLMs (e.g., LLaVA, Qwen-VL)                  |
| Libraries / Softwares | Pytorch, Tensorflow, Deepspeed, Huggingface, OpenCV                     |
| Developer Tools       | Git, Docker, Vim, VSCode, Cursor, Claude Code, PyCharm                  |